



ExoHabitAI – Final Deployment Documentation

1. Objective

The objective of ExoHabitAI is to deploy a machine learning–based web application that predicts the habitability of exoplanets based on planetary and stellar parameters.

The deployment integrates:

- Trained Machine Learning Model (XGBoost)
- Flask Backend API
- Frontend UI (HTML, CSS, JavaScript, Bootstrap)
- Production Server (Render Cloud Platform)

The final application is publicly accessible via a live URL.

2. Deployment Overview

The deployed system follows the architecture:

User → Frontend (Flask Templates)

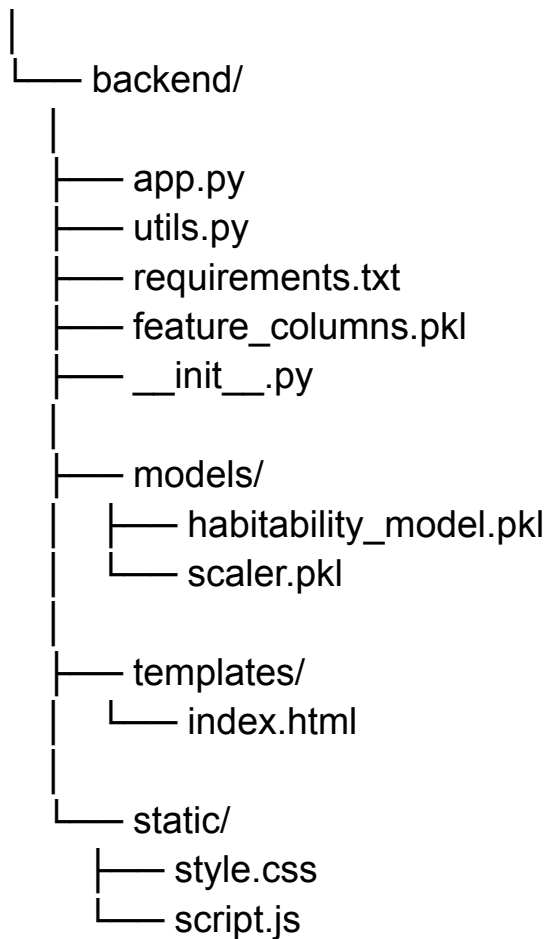
- Flask API (/predict)
- ML Model (XGBoost + StandardScaler)
- JSON Response
- UI Display

Deployment Platform Used:

- Render (Cloud Web Service)
-

3. Project Folder Structure (Final Deployed Structure)

ExoHabitAI/



Procfile (Root Directory):

web: gunicorn backend.app:app

4. Backend Preparation

Flask Application Includes:

- Flask app initialization
- CORS configuration
- Model loading using joblib
- StandardScaler loading
- Feature column alignment
- Routes:
 - `/` → Renders frontend
 - `/predict` → Returns habitability prediction
 - `/rank` → Ranks multiple planets

Production Configuration

```
if __name__ == "__main__":  
    app.run(debug=False)
```

The app is served in production using Gunicorn.

5. Machine Learning Model Details

- Algorithm Used: XGBoost Classifier
- Preprocessing:
 - StandardScaler
 - Feature alignment using `feature_columns.pkl`
- Output:
 - Binary Prediction (0 or 1)

- Confidence score using `predict_proba()`

Training Metrics Used:

- F1 Score
 - ROC-AUC Score
 - Classification Report
-

6. requirements.txt

The final requirements file includes:

```
flask
flask-cors
gunicorn
numpy
pandas
scikit-learn
xgboost
joblib
```

This ensures all dependencies are installed during cloud deployment.

7. Frontend Development

(According to Module 6 Frontend Guidelines)

UI Technologies Used:

- HTML5

- Bootstrap 5
- Custom CSS (Dark Blue Theme)
- JavaScript (Fetch API)

Features Implemented:

Input Fields:

- Planet Radius
- Planet Mass
- Orbital Period
- Equilibrium Temperature
- Star Spectral Type

Functionality:

- Prevents page reload on submit
- Sends JSON request to `/predict`
- Displays:
 - Habitability Status
 - Confidence Score
- Shows user-friendly error messages

UI Styling:

- Dark blue gradient theme
 - Responsive design
 - Status-based color highlighting (Green for habitable, Red for non-habitable)
-

8. Deployment on Render

Step 1: GitHub Integration

Project pushed to GitHub repository.

Link:-

<https://github.com/ARYAKRI/EXOHABIT-AI/tree/main>

Step 2: Create Web Service

- Environment: Python
- Build Command:

```
pip install -r backend/requirements.txt
```

- Start Command:

```
gunicorn backend.app:app
```

Step 3: Procfile Configuration

```
web: gunicorn backend.app:app
```

Step 4: Successful Deployment

Render provided public URL:

<https://exohabit-ai-chse.onrender.com>

9. Testing the Deployed Application

The following tests were performed:

- ✓ Home page loads correctly
- ✓ Input form works
- ✓ Prediction API returns correct output
- ✓ Confidence score displayed
- ✓ No crashes or server errors
- ✓ End-to-end pipeline verified

Flow Tested:

User Input → Frontend → Flask → ML Model → JSON → UI

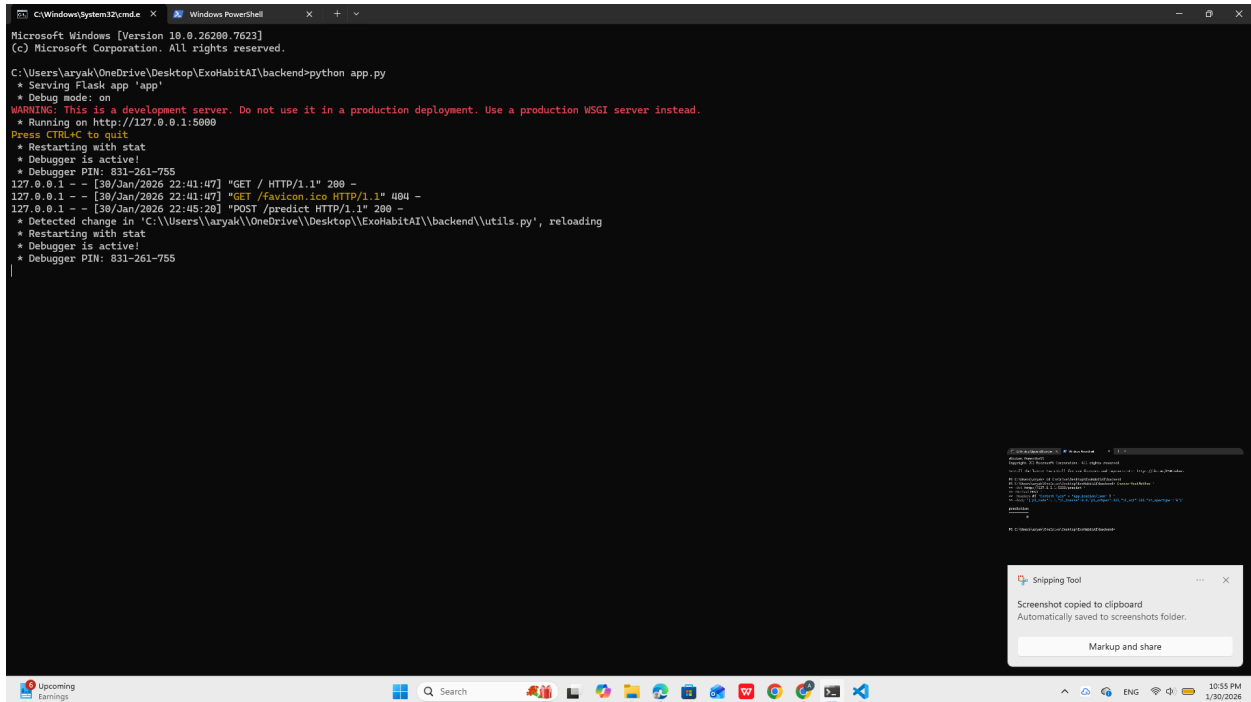
10. Common Issues Resolved During Deployment

Error	Cause	Solution
ModuleNotFoundError	Missing dependency	Updated requirements.txt
gunicorn not found	Not installed	Added to requirements.txt
CORS error	Missing flask-cors	Added flask-cors
utils module error	Import path issue	Ensured backend package structure
Method Not Allowed	Direct GET on /predict	Used POST request properly
Failed to fetch	Incorrect API URL	Used relative path <code>/predict</code>

11. Final Deliverables

- ✓ Live Deployment URL
- ✓ GitHub Repository
- ✓ Working Web Application
- ✓ Integrated ML + Backend + Frontend
- ✓ Deployment Documentation

Screenshots:-



```
Microsoft Windows [Version 10.0.26280.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\aryak\OneDrive\Desktop\ExoHabitAI\backend>python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 831-261-755
127.0.0.1 - - [30/Jan/2026 22:41:47] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [30/Jan/2026 22:41:47] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [30/Jan/2026 22:45:20] "POST /predict HTTP/1.1" 200 -
* Detected change in 'C:\Users\aryak\OneDrive\Desktop\ExoHabitAI\backend\utils.py', reloading
* Restarting with stat
* Debugger is active!
* Debugger PIN: 831-261-755
```

Snipping Tool

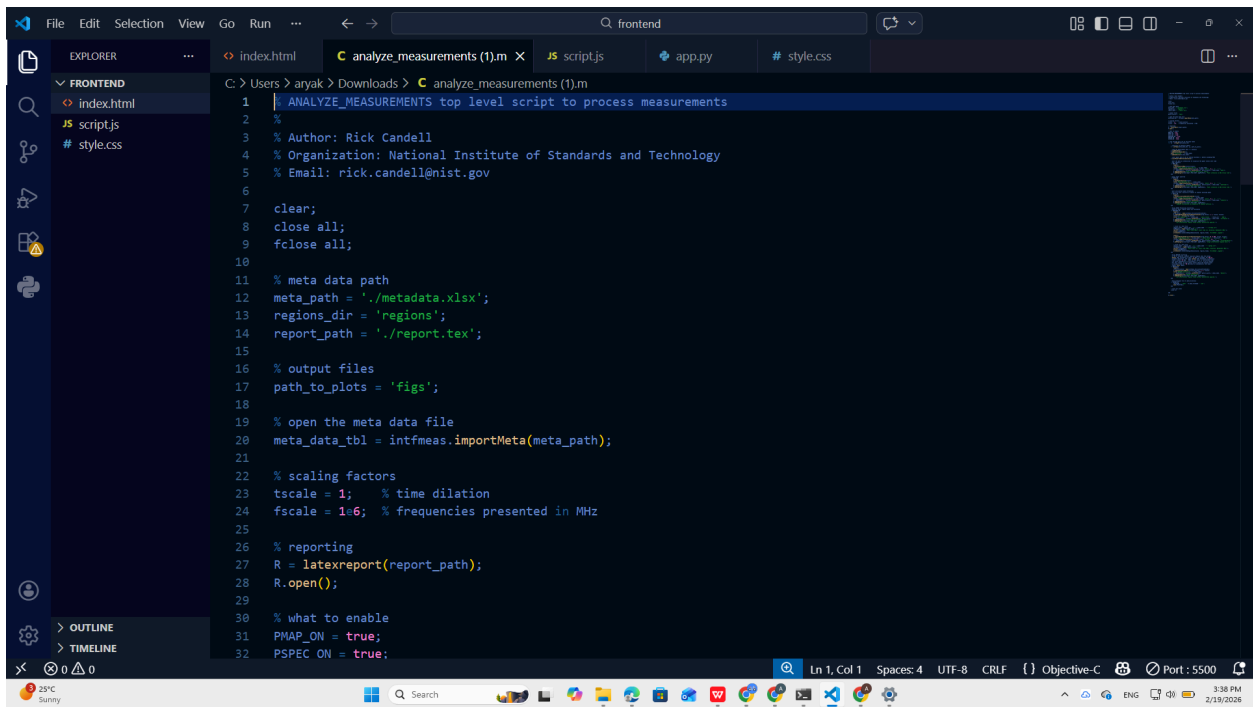
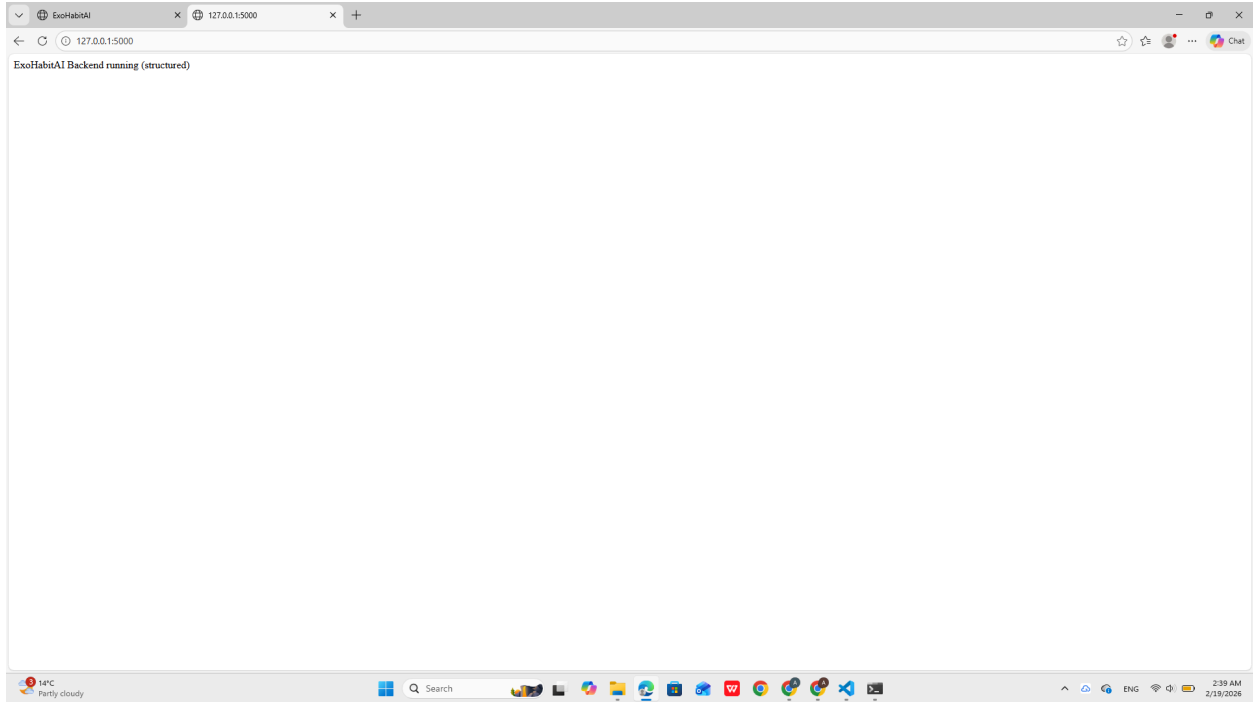
Screenshot copied to clipboard
Automatically saved to screenshots folder.

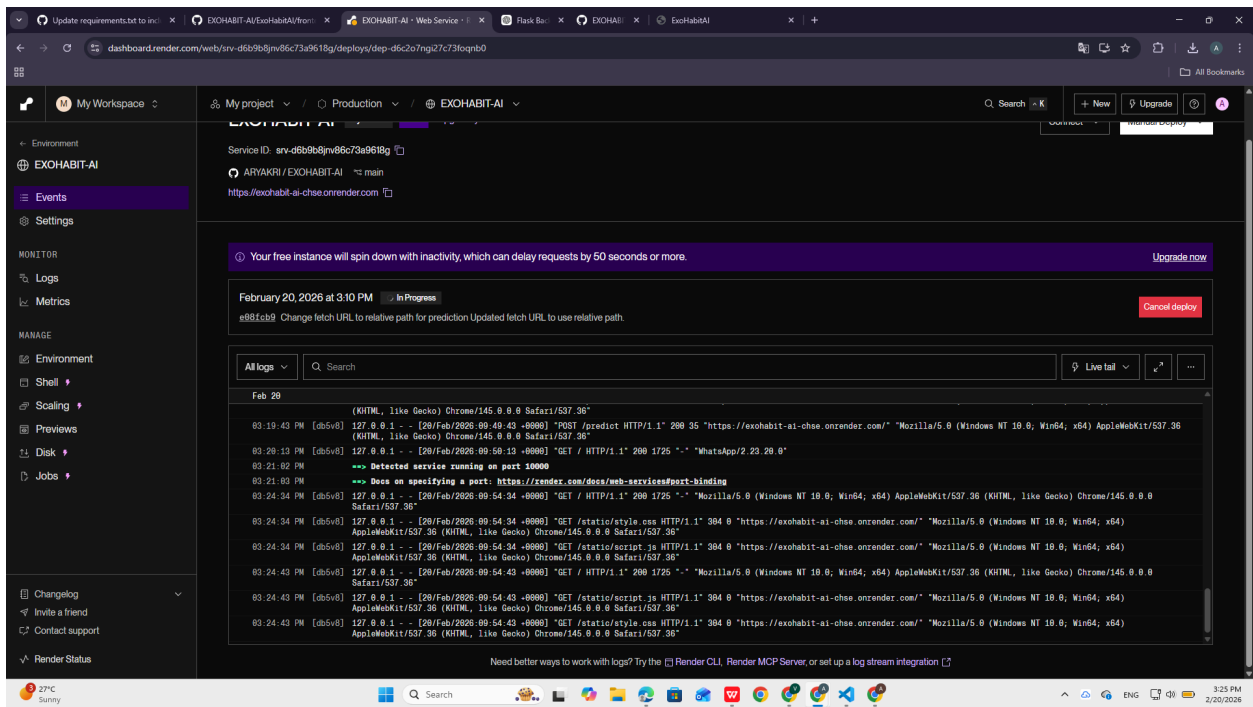
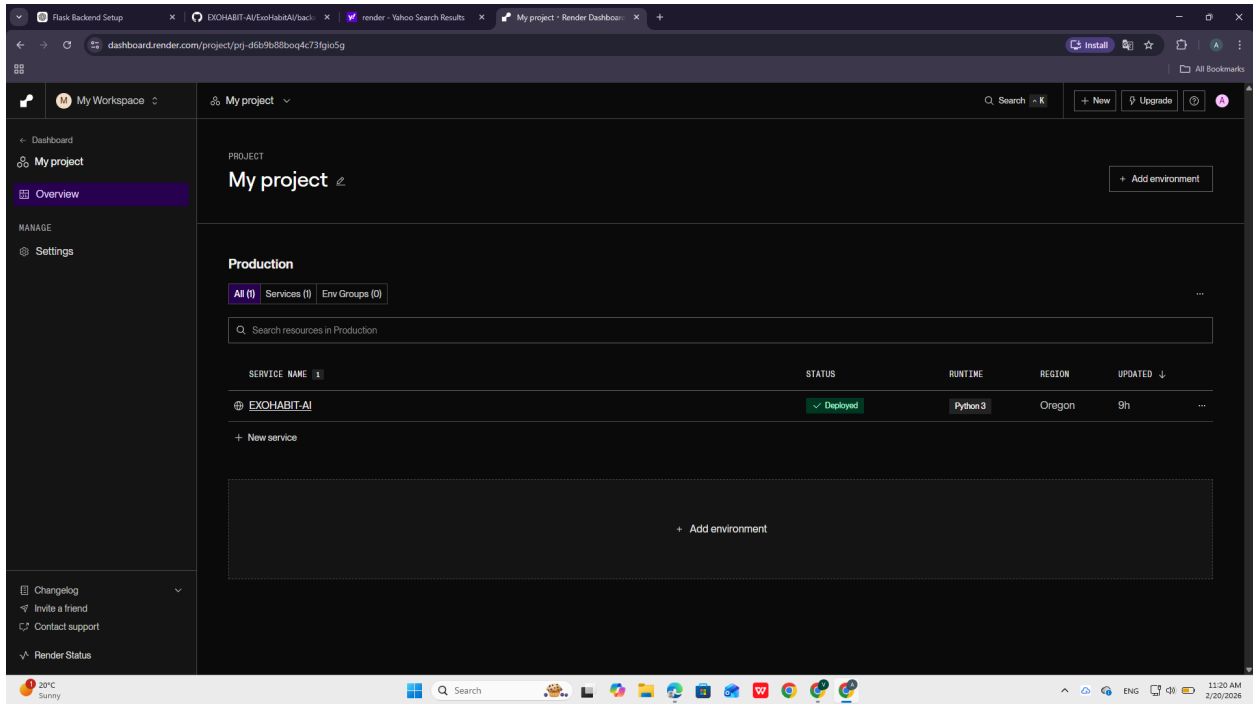
Markup and share

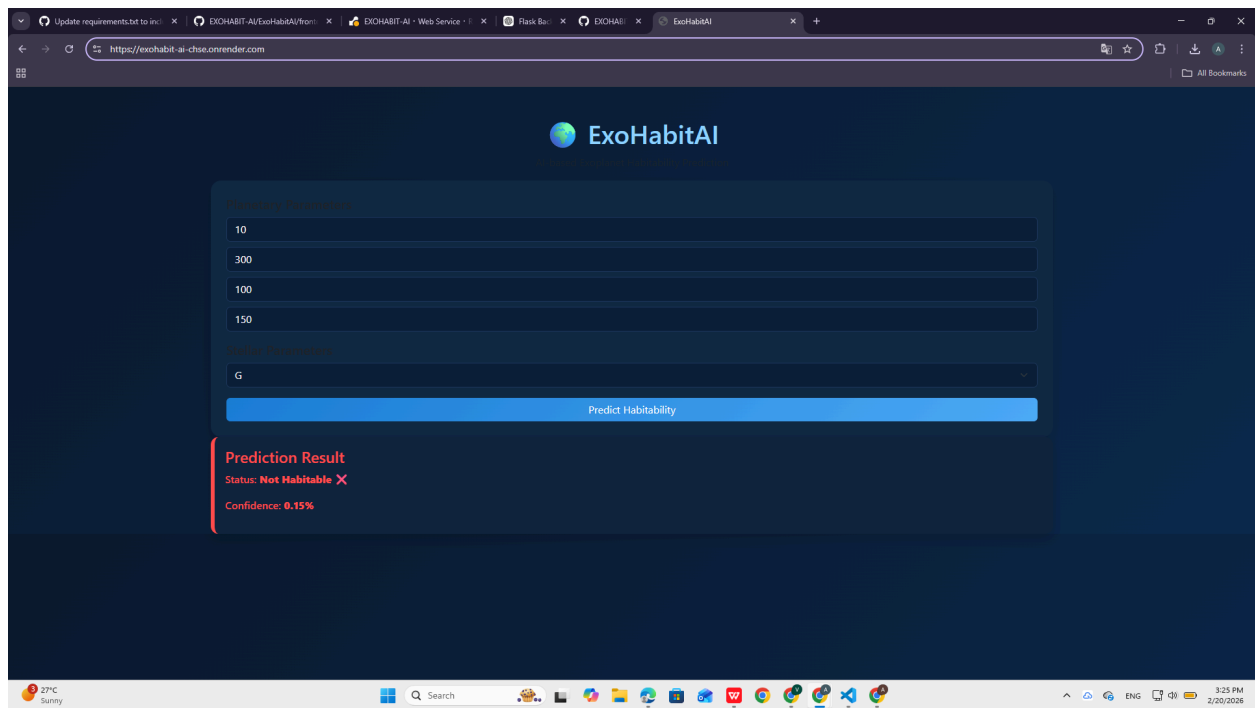
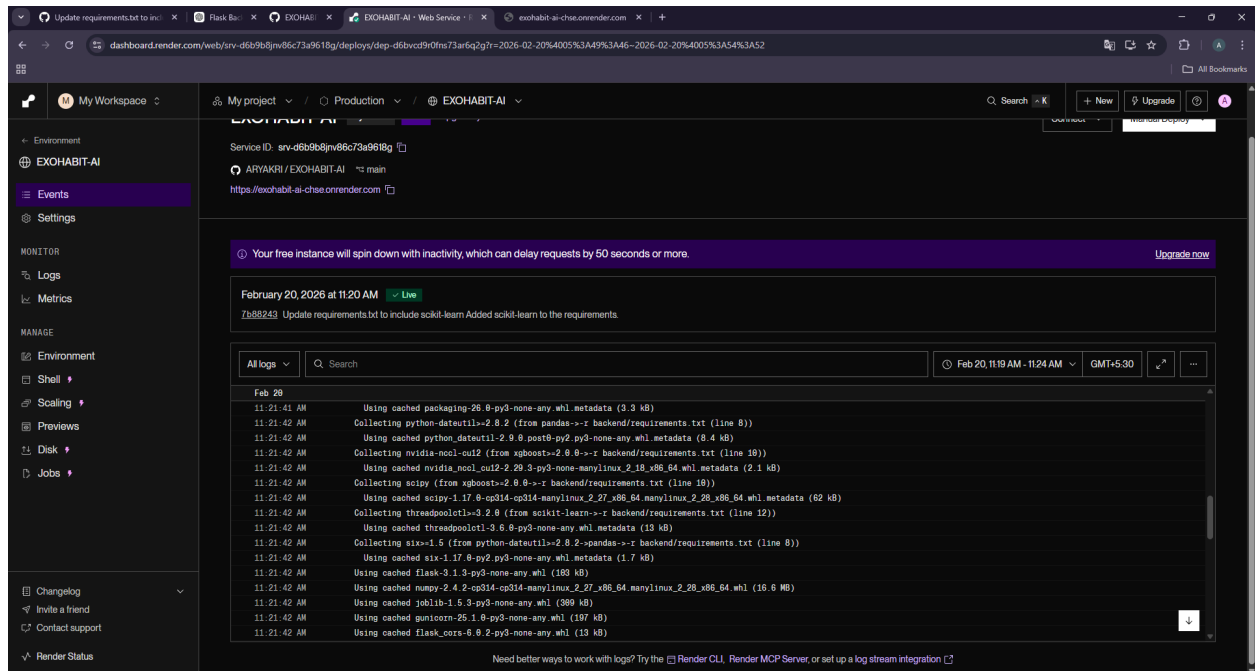
Upcoming Earnings

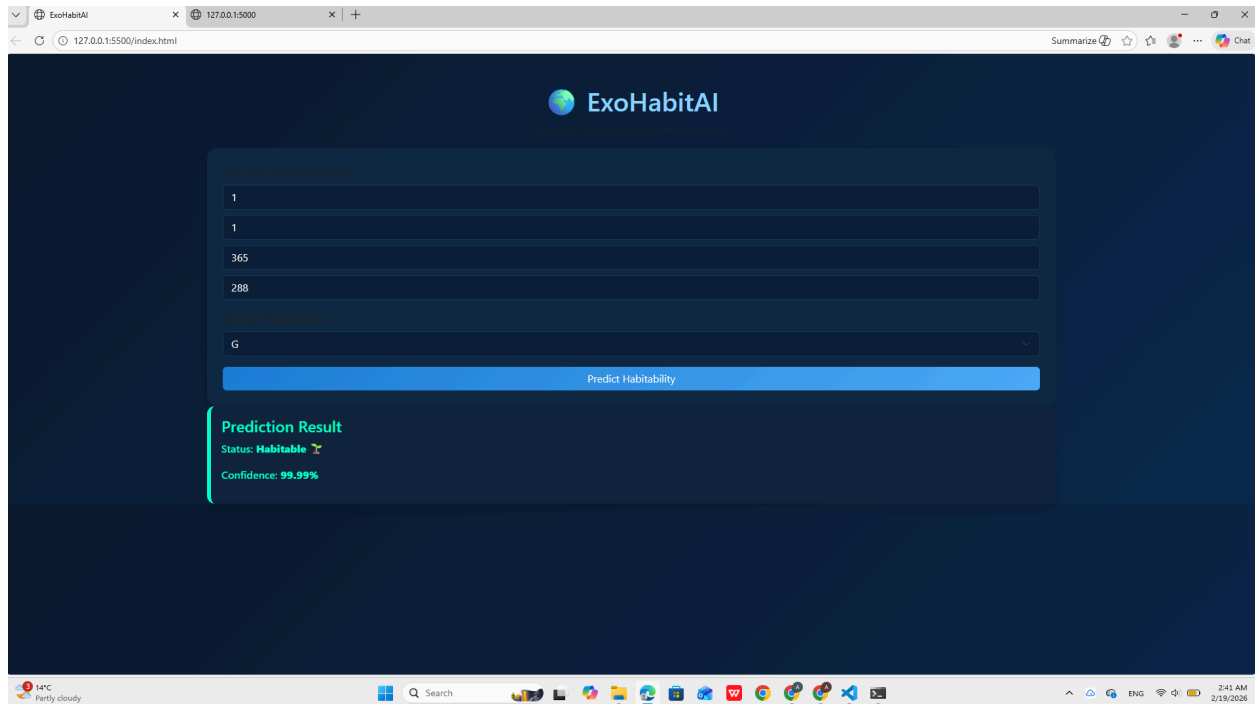
Search

ENG 10:55 PM 1/30/2026









12. Conclusion

ExoHabitAI was successfully developed and deployed as a full-stack machine learning web application.

The project demonstrates:

- ML model integration
- Backend API design
- Frontend development
- Cloud deployment
- Production debugging
- Real-world software engineering workflow

The application is publicly accessible and fully functional.

